

Explanation: Data and Model Summary

Pooled Predictor Distribution

The pooled predictor histogram takes all 40 predictor columns from the training set and plots their values together. It is not one variable by itself; it is a quick check of the overall scale and shape of the feature matrix.

In our data it is centered close to zero and looks roughly bell-shaped. That tells us the predictors are already on similar scales, but we still standardize inside cross-validation so every ridge and lasso fit treats coefficients fairly and avoids leakage.

Ridge and Lasso Formulas

Ridge minimizes: $(1/n) \sum_i (y_i - \beta_0 - x_i' \beta)^2 + \lambda \sum_j \beta_j^2$.

Lasso minimizes: $(1/n) \sum_i (y_i - \beta_0 - x_i' \beta)^2 + \alpha \sum_j |\beta_j|$.

The intercept β_0 is fitted but not penalized. Ridge shrinks all coefficients smoothly.

Lasso can set coefficients exactly to zero, so it also performs variable selection.

Reported Quantities

- GD vs closed-form RMSE = $2.25e-11$. This is the root mean squared difference between gradient-descent ridge test predictions and closed-form ridge test predictions. A value this small means GD converged to the same ridge solution.
- Ridge test pred mean = -1.059. This is the average predicted y value across the 100 hidden test rows from the final ridge model.
- Lasso test pred mean = -0.924. This is the average predicted y value across the 100 hidden test rows from the final lasso model.

Intuition: ridge and lasso are both linear prediction models, but ridge prioritizes stable shrinkage while lasso prioritizes sparse variable selection.

Explanation: Bottom Three Plots on Slide 3

4. Bootstrap Lasso Stability

- Each bar is one predictor, sorted by how often lasso selected it across 200 bootstrap refits.
- Purple bars were selected by the full-data lasso model; gray bars were not selected by the full-data lasso model.
- The red dashed 50% frequency line means the variable was selected in half of the bootstrap samples. Variables far above 50% are more stable; variables near 50% are borderline and sensitive to resampling.

5. Gradient Descent Convergence

- In our solution, gradient descent is used only for the ridge part. First, closed-form ridge plus 5-fold CV chooses λ . Then we run gradient descent with that same λ as an independent check that iterative optimization reaches the same ridge coefficients.
- It does not mix with lasso. Lasso is fit separately with sklearn LassoCV, because libraries are allowed for the lasso tasks. Ridge and lasso predictions are both reported, but their fitting procedures are separate.
- β_{GD} is the ridge coefficient vector found by iterative gradient descent after a given number of steps.
- β_{closed} is the ridge coefficient vector found directly from the closed-form matrix solution.
- The green curve plots $\|\beta_{GD} - \beta_{closed}\|$. Its drop toward zero shows that gradient descent is approaching the same optimum as the closed-form ridge formula.
- The dashed objective curve shows the penalized ridge objective decreasing, which is what gradient descent is designed to do.

6. Penalty Geometry

- Here β_1 and β_2 are two example regression coefficients. The plot shows a two-coefficient cartoon so we can see the penalty shapes; the real assignment has 40 beta coefficients.
- The ridge L2 circle represents the shape of the ridge penalty $\beta_1^2 + \beta_2^2$. Because the boundary is smooth, ridge usually shrinks coefficients but does not set them exactly to zero.
- The lasso L1 diamond represents the shape of the lasso penalty $|\beta_1| + |\beta_2|$. Its sharp corners make it likely that the optimum lands on an axis, giving an exactly zero coefficient.
- The lasso corner marker illustrates this sparse solution behavior. The ridge smooth shrink marker illustrates ridge's tendency to keep both coefficients nonzero but smaller.